HR ATTRITION ANALYSIS

End-to-End Data Analytics Project Report

Python | SQL | Power BI

Project Domain	Human Resources Analytics
Prepared by	Padmanaban L
Tools	Python, Pandas, Matplotlib, Seaborn, SQL, Power BI
Status	Completed

This report documents the workflow, analysis, dashboard, and findings of an HR employee attrition analytics project. It combines Python-based data inspection and exploratory analysis, SQL business queries, and a three-page Power BI dashboard.

Executive Summary

The project explores employee attrition patterns and workforce characteristics to support evidence-informed HR review. Python was used to inspect the dataset, check data quality, calculate attrition distributions, compare attrition by overtime and job satisfaction, and export a cleaned CSV. SQL was used to formulate business questions across workforce size, exits, departments, job roles, tenure, income, demographics, experience, education, performance, training, travel, and promotion-related indicators. Power BI presents the results through interactive KPI cards, slicers, charts, and employee detail tables.

The dashboard displays 7K employees, 475 employees left, an attrition rate of 6.79%, and average monthly income of 99.60K. It also reports average employee age of 40.12, average training sessions of 3.00, average total experience of 20.03, and average distance from home of 15.34. Values are reproduced from the supplied dashboard screenshots.

1. Business Problem

Employee turnover can increase recruitment and onboarding effort, disrupt teams, and affect continuity. HR stakeholders need a consolidated view of workforce composition and attrition patterns, with the ability to investigate differences across departments, job roles, and employee experience indicators.

2. Project Objectives

- Measure workforce size, employee exits, and attrition rate.
- Explore attrition across departments and job roles.
- Inspect associations between attrition and overtime, job satisfaction, and work-life balance.
- Summarize workforce demographics, experience, training, business travel, education, and income.
- Build an interactive Power BI dashboard for exploration.
- Translate descriptive findings into practical areas for HR follow-up.

3. Dataset Overview

The project code references HR_Analytics.csv and the SQL queries use the table hr_analytics_cleaned. Fields referenced in the supplied scripts include EmployeeID, Attrition, Department, JobRole, Age, Gender, MonthlyIncome, OverTime, WorkLifeBalance, JobSatisfaction, MaritalStatus, JobLevel, BusinessTravel, Education, TrainingTimesLastYear, TotalWorkingYears, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, NumCompaniesWorked, PerformanceRating, and DistanceFromHome.

The raw dataset itself was not included with the uploaded scripts; therefore, exact source provenance, row-level validation results, and full query result tables are not asserted beyond the supplied dashboard values.

4. Tools & Technologies

Tool	Project use
Python / Pandas	Data loading, inspection, data-quality checks, crosstab analysis, export
Matplotlib / Seaborn	Visualizing attrition distribution and attrition rates
SQL	Answering business questions with filtering, aggregation, grouping, ordering, and subqueries
Power BI / Power Query / DAX	Data preparation, KPI measures, interactive dashboard and visuals
Google Colab	Notebook execution and cleaned-file download

5. Methodology

1. Load and inspect the employee dataset.
2. Check data types, missing values, duplicate rows, and categorical values.
3. Use Python to calculate overall attrition and compare attrition percentages by overtime and job satisfaction.
4. Export the prepared dataset as HR_Analytics_Cleaned.csv for downstream use.
5. Use SQL queries to answer business questions through COUNT, AVG, CASE, GROUP BY, ORDER BY, WHERE, and a scalar subquery.
6. Build Power BI pages with KPI cards, filters, charts, and detail table; review displayed measures against dashboard screenshots.

6. Python Analysis

The uploaded Python script is a notebook-style exploratory analysis. The steps below reflect the code supplied.

1. Load the dataset: The script imports pandas and reads HR_Analytics.csv into a DataFrame named df.

2. Inspect structure and summary: The script uses head(), info(), describe(), describe(include='object'), describe(include='all'), dtypes, and value counts to understand the dataset.

3. Check data quality: The script checks missing values with isnull().sum(), duplicate rows with duplicated().sum(), and validates categorical values using value_counts() and unique().

4. Analyze overall attrition: The script calculates Attrition value counts and normalized percentages, then visualizes the employee attrition distribution with a Seaborn count plot.

5. Examine overtime: A normalized crosstab compares Attrition by OverTime; the attrition percentage for each overtime category is plotted.

6. Examine job satisfaction: A normalized crosstab compares Attrition by JobSatisfaction rating and plots the attrition percentage for each rating.

7. Export prepared data: The script writes the dataset to HR_Analytics_Cleaned.csv and downloads it from Google Colab.

Python code highlights

```
attrition_rate = df['Attrition'].value_counts(normalize=True) * 100
overtime_attrition = pd.crosstab(df['OverTime'], df['Attrition'], normalize='index') * 100
satisfaction_attrition = pd.crosstab(df['JobSatisfaction'], df['Attrition'], normalize='index') * 100
df.to_csv('HR_Analytics_Cleaned.csv', index=False)
```

7. SQL Analysis

The supplied SQL script contains 22 commented business questions. Queries use the hr_analytics_cleaned table and demonstrate aggregation, conditional counting, filtering, grouping, sorting, and a subquery.

7. What is the total number of employees in the organization?
8. How many employees have exited the organization?
9. What percentage of the workforce has experienced attrition?
10. How is the workforce distributed across different departments?
11. Which job roles have the highest employee attrition?
12. Does employee tenure differ between employees who stayed and those who left?
13. Is monthly income associated with employee attrition?
14. What is the average age of employees in each department?

15. Which department has the highest number of employees aged above 50?
16. Which job roles have employees who have spent the longest time in their current role?
17. How many employees have more than 20 years of total work experience?
18. What is the average number of companies previously worked for by employees in each department?
19. What is the average education level across departments?
20. Which job roles have the highest average performance rating?
21. What is the average number of training sessions received by employees at each job level?
22. Which employees have received no training in the last year?
23. Which employees travel frequently for business and have low job satisfaction?
24. Which marital-status group has the highest average monthly income?
25. Which employees have changed roles less frequently than expected for their tenure?
26. Which employees have worked in more companies than the average employee?
27. Which employees have not received a promotion despite having at least 10 years at the company?
28. Which education levels have the highest average monthly income?

SQL techniques used

- COUNT(*) and AVG() for counts and averages.
- CASE WHEN to calculate the attrition numerator.
- GROUP BY to compare departments, roles, and employee segments.
- WHERE filters for targeted employee groups.
- ORDER BY to rank results.
- A scalar subquery to compare an employee's number of previous companies with the overall average.

Note: SQL query outputs were not included in the uploaded .sql file. This report describes the questions and methods; query result values should be added after execution against the project database.

8. Power BI Dashboard

The supplied dashboard contains three pages. Slicers and visuals support interactive exploration; screenshots are included below.

Page 1 — HR Attrition Analysis Dashboard

Overview page: department and gender filters; KPI cards for employee count, employees left, attrition rate, and average monthly income; employees left by job role; employee count versus employees left by department; attrition distribution; and gender distribution.

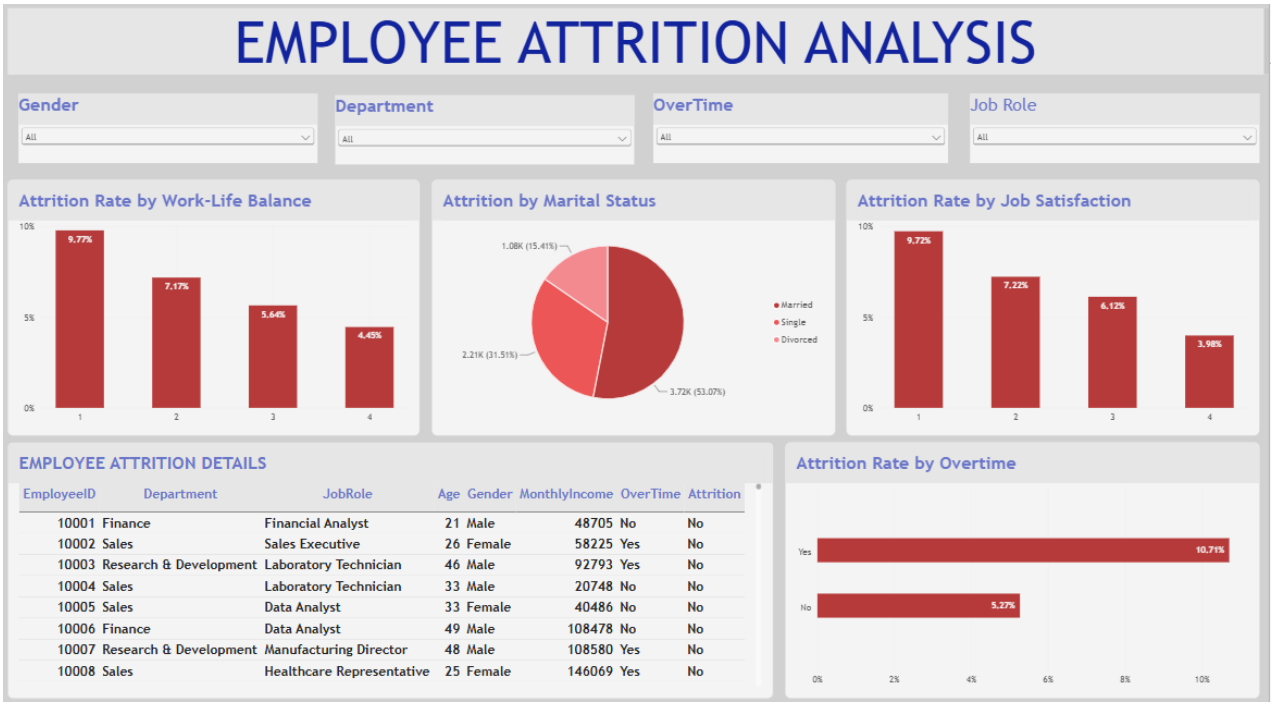


Figure 1. HR Attrition Analysis Dashboard.

Page 2 — Employee Attrition Analysis

Detailed attrition page: gender, department, overtime, and job-role filters; attrition rate by work-life balance and job satisfaction; attrition by marital status; attrition rate by overtime; and an employee details table.

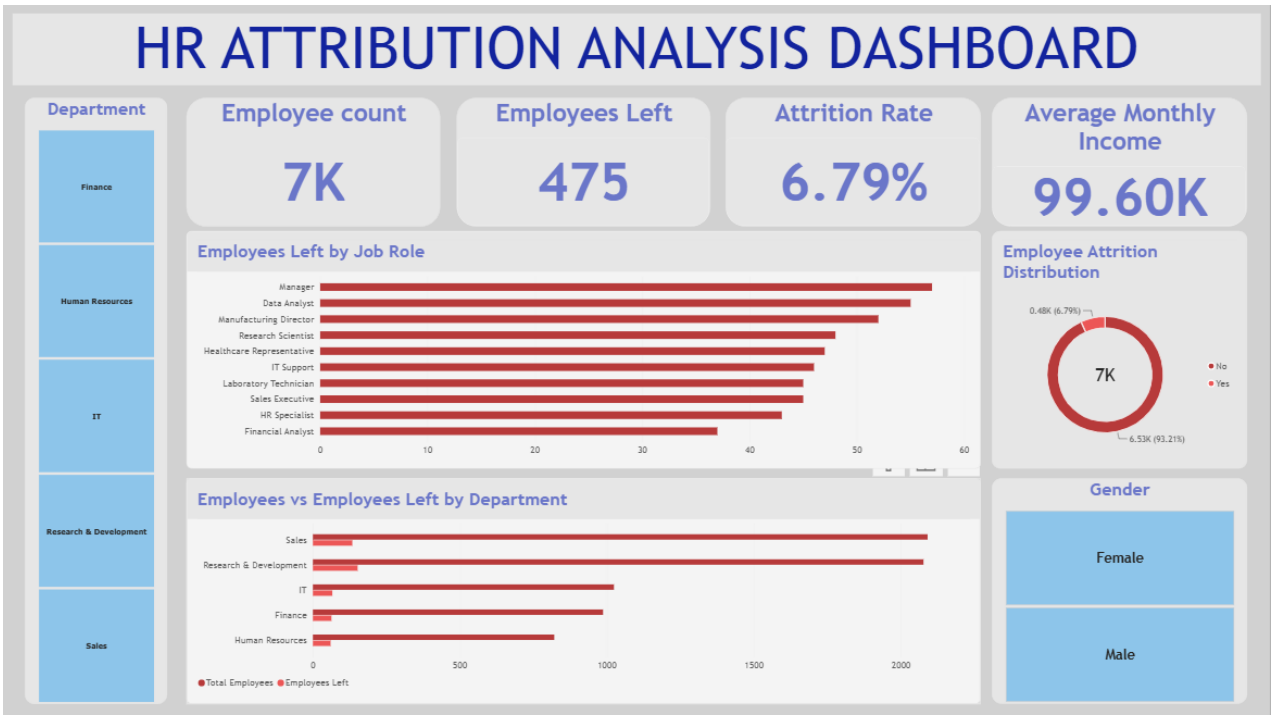


Figure 2. Employee Attrition Analysis.

Page 3 — HR Workforce Insights

Workforce profile page: average age, training sessions, total experience, distance from home, average experience by job level, business travel, education, age distribution, and total monthly salary by job role.



Figure 3. HR Workforce Insights.

9. KPI Summary

Metric	Dashboard value
Employee count	7K (rounded display)
Employees left	475
Attrition rate	6.79%
Average monthly income	99.60K
Average employee age	40.12
Average training sessions	3.00
Average total experience	20.03
Average distance from home	15.34

10. Key Observations

- The dashboard reports 475 employees left and an overall attrition rate of 6.79%.
- Overtime is associated with a higher displayed attrition rate: 10.71% for employees marked Yes versus 5.27% for No.
- Displayed attrition rates decrease as work-life balance rating increases: 9.77% at rating 1, 7.17% at 2, 5.64% at 3, and 4.45% at 4.
- Displayed attrition rates decrease as job satisfaction rating increases: 9.72% at rating 1, 7.22% at 2, 6.12% at 3, and 3.98% at 4.
- Job-role and department visuals provide a way to identify where exits are concentrated; exact rankings should be verified from the underlying data or report.

Interpretation: These are descriptive associations shown in the dashboard. They do not establish that overtime, satisfaction, or work-life balance causes attrition.

11. Business Recommendations

- Review overtime workload, staffing levels, and manager practices in groups with elevated attrition.
- Use employee feedback and engagement surveys to investigate low work-life balance and job satisfaction scores.
- Develop targeted retention actions for departments and job roles with higher employee exits.
- Review compensation, career progression, and internal mobility using role- and level-specific analysis.
- Monitor attrition KPIs over time and validate metric definitions with HR stakeholders before operational decisions.

12. Limitations

- The raw CSV and executed SQL result outputs were not supplied with the code files.
- The dashboard employee count is shown as rounded 7K; the exact row count should be confirmed from the dataset.
- Some dashboard labels require validation against the source measure definitions before making detailed claims.
- The analysis is descriptive; causal conclusions require additional statistical testing and contextual HR investigation.

13. Conclusion

This project demonstrates an end-to-end analytics workflow for HR attrition: Python for data inspection and exploratory analysis, SQL for structured business-question analysis, and Power BI for interactive reporting. The combined workflow turns employee data into accessible indicators that can guide further HR investigation and retention planning.

Appendix A — SQL query file

The complete SQL script is maintained separately as hr data analysis(1).sql. It includes the commented business questions and executable queries. The script should be stored alongside this report in the project repository.

Appendix B — Python script

The Python script is maintained separately as hr_analysis(2).py. It reads HR_Analytics.csv and exports HR_Analytics_Cleaned.csv from Google Colab.